

IB Math IA Guide - Math IA Shenanigans That No One Will Tell Yeh!

Ace Your IB Math IA with our ultimate guide for 2023! Get top marks and ace your IA with ease. Discover proven tips, tricks and strategies to nail your Math IA today!

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IB Math students will tell you how they're always on the edge of their seats for some help, but IB

Math IA takes that anxiety to an entirely different level. The reality is far from frightening;

nonetheless, IB Math IA can be handled well with a unique IB Math IA topic in hand and lots of coffee!

But does that guarantee a dependable 7?

It takes more than just a perfect IB Math IA topic to ace.

How's that, you'd ask.

From researching several IB Math IA examples to planning the mathematical working of your exploration, your IB Math IA structure will get you into trouble if you don't give it the time it demands. With all the varied content available in bulk online, the process is bound to become anything but easy.

But worry not!

You are at the right place - The Ultimate Guide to IB Math IA!

This article covers IB Math IA rubrics, process key pointers, the structure of the investigation, and interesting IB Math IA topics that will stimulate your mind and help you begin your exploration!

You should also know about the updated course structure of IB Mathematics. Students are allowed to opt for any one of the following four courses in Math:

- [Analysis and Approaches \(AA HL\) - Higher Level](#)
- [Analysis and Approaches \(AA SL\) - Standard Level](#)
- [Applications and Interpretation \(AI HL\) - Higher Level](#)
- [Applications and Interpretation \(AI SL\) - Standard Level](#)

For more information about choosing your course, check out our latest article: [Confused about IB Math AA & IB Math AI?](#)

Also, here's a great surprise for all students!

[Patrick Jones](#), the creator of PatrickJMT Math Videos, acknowledged as the best Math teacher globally with over 1.2 million subscribers on YouTube, has gotten on board with our team at Nail IB! How great is that! He is already working on creating an entire Nail IB video course, and it will prove to be a wholesome guide for you as you tread on your IB Math journey! You should check out his excellent, world-renowned content [here](#)!

Before moving any further, we insist you check out our Free IB Resources for [IB Mathematics SL and IB Mathematics HL](#). These are specially assembled for your benefit and will surely assist you on your IB Math journey!

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For an absolute hold on IB Math, check out our premium notes designed and curated specially for you, be it [IB Mathematics SL](#) or [IB Mathematics HL](#). These bundles are not just limited to messages but offer past year papers and How-to Guides for Extended Essays, Internal Assessments, and more; examples included! You're in for a smooth ride with these by your side;)

You can also stream our webinar on [How to Write an IB Math Internal Assessment in under 30 minutes](#) and hear directly from a recent IB graduate to understand the fundamental pointers and some fantastic hacks to lay the foundation of your IB Math IA. Getting the proper guidance ensures you a 7 in the subject you have feared for too long. Click [here](#) to watch it now!

First things first, let's understand the criteria. Unless we acknowledge the requirements against which our exploration is scored, it'll be equivalent to a shot in the dark. The conditions, irrespective of whether you opt for SL, HL (AA or AI), are as follows:

Criterion A - Mathematical Presentation: [Levels- 0, 1, 2, 3, 4]

Assessment is done on the conciseness, brevity, and clarity/coherence of your investigation. The proper structure must be given to your IA. As per IB guidelines,

a coherent exploration is,

- Logically developed
- Easy to follow and,
- Meets the Aim.

Also, a well-organized exploration,

- Includes Introduction
- Describes the Aim of the investigation and
- Has a Conclusion

Criterion B - Mathematical Communication: [Levels- 0, 1, 2, 3, 4]

Assessment is done on the appropriateness of the mathematical terminology, notation, and symbols used to progress the exploration. Marks and notes should be correctly used as are used in IB textbooks. For example, x^2 should not be written as $x^{\wedge}2$.

If used, different mathematical representation tools such as tables, graphs, and diagrams must be relevant to the working and be commented on/explained well. Avoid inconsistent use of Mathematical terminology. Applying ICT Tools(for example- [GeoGebra and Desmos](#)) should be made wisely. For Calculations, Graphic Display Calculators can also be used, but that doesn't outdo math formulas' importance.

Criterion C - Personal Engagement: [Levels - 0, 1, 2, 3]

Assessment is done on the personal involvement shown. The sure shot way to ensure Engagement is, first and foremost, by going ahead with a topic that interests you (something unique or that affects real-life situations). Personal Engagement is seen throughout the exploration by:

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and creativity showed by the student

and your own

- Investigating the idea from varied perspectives
- Exploring different possibilities

Avoid portraying superficial interest. Opportunities for demonstration of personal Engagement should be noticed.

Criterion D - Reflection: (Levels - 0, 1, 2, 3)

Assessment is done on the evaluation and analysis of the investigation. Mentioning the significance of your exploration results, discussing possible limitations, and justifying why you chose the procedure you did can portray a fair reflection of the IA. Merely explaining your results will get you only a score of 1 out of 3. According to IB guidelines, a review should be meaningful and critical.

A meaningful reflection includes

- Considering limitations in the work
- Comparing different Mathematical Approaches
- Linking to the Aim
- Commenting on the Learning

A critical reflection entails,

- Considering **What Next**
- Discussing the implications of the results
- Discussing the strengths and weaknesses of the approaches
- Considering different perspectives

Reflection is an analysis of the student's work, seen throughout the exploration, not just the Conclusion.

Criterion E - Use Of Mathematics: (Levels - 0, 1, 2, 3, 4, 5, 6)

Assessment is done on the implementation of Mathematics in the IA. It is essential to understand that the Math used should be on par with the course, nothing too simple and nothing you need help understanding. Also, the Mathematics used should be fully understood and engrained by you. Unfamiliar Mathematics, if used, should be explained well by giving personal examples. As per IB guidelines, students are expected to produce work that is,

- **Commensurate** with the level of the course[should either be part of the syllabus at a similar level or slightly beyond]
- **Relevant** Mathematics used means Math which supports the development of the exploration towards the completion of its Aim

To score higher levels in Criterion E, it is crucial to understand the meaning of the following terms:

Precise Mathematics is error-free and uses an appropriate level of accuracy at all times

Math should be commensurate with the HL syllabus

- **Rigour** involves clarity of logic and language when making Mathematical arguments and calculations.

[Here](#) is a fully annotated sample IB Math IA, going through which you can gain a lot of insight [**Read the annotations properly**].

Another example, [the Breaking the Code](#) investigation[fully annotated]- is given here for IB Mathematics HL. The IA explores encryption and decryption in the context of Mathematics.

- Going through the report, we see that the document needs more structure in the beginning since the Introduction does not mention the Rationale or Aim. You don't want to be committing such a blunder.
- Moving further, we see that the body encompasses Math, which is well explained, thereby excelling on criterion E when graded on the SL scale. To excel in HL, the math should have been more rigorous than descriptive.
- Toward the end, the report needs to include a Reflection on the results obtained, which doesn't fare well for the IA. Potential implications of the topic need to be included, and the Conclusion seems bland.

Another interesting annotated sample- for IB Mathematics SL- Regularisation of Irregular Verbs: When can I use the words swimming and know correctly? is given for reference [here](#). Understand which key points have been missed and which have been taken care of. The more you go through sample IAs, the better your chances of preparing an investigation that'll be scored well.

Like any other exploration, your IB Math IA expects you to give it a fair shot of effort and interest. If you see it as a burden, it will undoubtedly become one. Equally important is to draw an analogy between the topic of your choice and the math involved. Taking care of these points in general, let's understand all that goes into making one's IB Math IA [a brief outline]:

- **Brainstorm for a Math IA topic you're curious about:**
 - It can have a personal story attached; mention it in your Introduction. If not, explain how the topic underhand impacts real-world situations and motivates you to land on it. Your passion for the IA idea shows in your work, and you don't want to be doing your investigation just for the sake of it.
- **Proper research is expected on your end:**
 - If you're curious about [Fibonacci numbers](#), [the Golden ratio](#), and nature alike, try looking for relationships among them on the Internet. This will entail going through many research papers, publications, and journals and finally settling on mathematical findings and proofs that will help you investigate that particular something you wish to explore. For example, if you want to study how [Traffic Jams have math running in the background](#), research it in detail since Traffic snarls are an imminent pain for us all. Only you will have to pick up parts you think are relevant and understandable to

- The cycle of Inquiry, Action, and Reflection in learning is vital. Learning the implication of [Plagiarism, Collusion, and Duplication of Work](#) is essential to keep one's IA transparent and impressive.

- **Referencing:**

- Besides citing references in the bibliography section, ensure you include it in the body as a footnote or in the exploration itself. Citing credible sources shows how transparent your work is and helps examiners cross-check for correctness. Acknowledging the author's work is essential to the IA-making process.

With this, let's discuss the Structure/Layout of the Investigation. There are numerous guidelines available all over the Internet. Regarding the IA length, though you should keep 6 -12 pages as the prescribed length, your focus should be on including all that pertains to your idea and ruling out everything that's not. So don't set out with a mindset to refrain from exceeding six pages; set out to include everything you know needs to be. Similarly, it is advised to make sure the Math used is suitable for SL and HL levels.

Without any further adieu, let's highlight what the layout of the IA should look like:

Introduction

- The Introduction of your exploration can be broken down into three essential parts -
 - **Rationale**
 - Sets the background of your exploration and gives an argument for your topic choice. Your Rationale tells why you chose so and so topic.
 - **Aim of the idea/topic**
 - This is where you define your investigation's objective and tell what you wish to achieve with this idea.
 - **Plan of Action (stating what area of the math curriculum your exploration focuses on)**
 - Let's say, for instance, you have opted for Math HL; for a simple mathematical investigation that scores six on the Math SL grade scale, you might end up with a mere four on the HL. The difficulty of your opted Math subject should reflect in your Internal Assessment. It would help if you also outlined the areas of mathematics you will cover in your investigation.

Body Of Your Exploration

- Elaborate on the method you used for the exploration and justify why you chose to proceed with that particular method.
- Use relevant mathematical tools like labelled graphs, charts, etc., for your mathematical work and explain them in the IA context.

Conclusion

- State your results relating them to the Aim of your Internal Assessment. The significance and impact of these results should be highlighted, as well. In addition, briefly tell how the exploration was helpful to you and all you have gained from it. Possibilities of extension

The bibliography is for you to cite the sources used by you in the exploration. I suggest you use the [Citation Machine](#) for additional guidance in the

Now that we're comfortable with the IB Math IA structure let's look at some interesting IB Math IA topics that will get your creative juices flowing and help kickstart your Math IA journey today!

- Simulating models to study and forecast weather patterns. [You could come up with a personal account that led you to land on something like this]
- Exploring the different probabilities associated with a game of your choice; for example, Solitaire(if you're a game buff).
- Investigating the [Math associated with the Global Positioning System\(GPS\)](#) and the intricacies of the technology involved.
- Exploring [Fermat's little theorem](#) or [Goldbach's conjecture](#)(one of the most significant unsolved problems in Mathematics).
- Finding the volume and surface area of an egg, apple, mango or any other real-world object using Calculus's power [Simulation could be used]. A good IA on **modelling** manages to score an easy 15-16 marks out of 20.
- Investigating the structural designs of bridges that prevent collapse under loading.
- Studying complex roots graphically.
- Exploring how guitar frets are arranged in Pythagoras Ratios.
- Comparing which will prove beneficial: lump-sum payment of a lottery prize or fee done in instalments?
- Understanding how ISBN codes and Credit Card Codes can be cracked.

And that's a wrap!

Just like any other IA, IB Mathematics IA needs to be started early so that you don't end up compiling just anything at the last minute!

Give it the time it needs, and it will surely pay off. It might seem heavy, but once you decide to pursue an idea of your liking, there will be no turning back! Keep in mind the essential pointers and win the battle courageously!

Want some A-quality guidance? Look no further; at Nail IB, we have assembled premium content for you to ace your IBs, and you should check out our resources for a smooth IB experience.

Click [here](#) for top-notch IB resources or to assess how your prep is going! Our exclusive Nail IB course, created by Patrick Jones, will be out soon too, so stay tuned, as there is no way you would want to miss the holy grail every Math IB student wants!!

This article will serve as a solid foundation for your Math IB Internal Assessment.

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